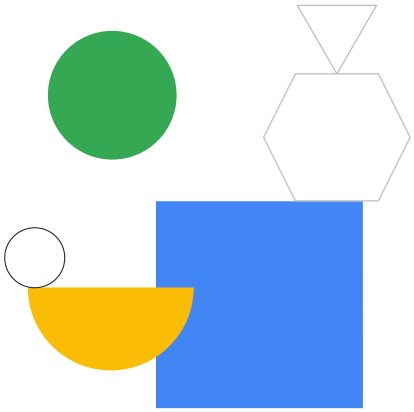
Google Cloud

Upgrade



In this module, you will learn how to upgrade your Apigee hybrid installation to a new release of the software distribution.

We will also discuss rollback procedures so you can roll back the Apigee hybrid installation to a previous version.

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Agenda

01 Upgrade Process

02 Rollback Process

03 Quiz

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In this lecture, we will review the process to upgrade Apigee hybrid to a newer version of the software distribution.

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Upgrade prerequisites

Before performing an upgrade:

- Review official documentation and release notes for specific considerations related to upgrade path and target version.

  Apigee hybrid release notes
- Verify that a supported version of Helm and cert-manager is available for installation.
- Back up your hybrid installation (recommended)
- Back up your hybrid runtime Cassandra database (recommended).
- Follow steps to enable analytics publisher access.

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Before performing an upgrade, you should always review the release notes for the Apigee hybrid release version you are upgrading to.

The release notes contains important information about the new features that are released, any features that are in beta, bug fixes, any required configuration file changes, and any updates to supporting software platforms.

It is also recommended to back up your hybrid runtime installation and Cassandra database following the instructions in the Apigee hybrid documentation.

Apigee hybrid release notes
Enable analytics publisher access

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Kubernetes platform upgrade

- Upgrading your version of Apigee hybrid requires a supported version of the Kubernetes platform, as documented in the release notes.
- Follow the platform documentation to upgrade Kubernetes to the required version for Apigee hybrid.

Apigee hybrid: supported platforms and versions

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Upgrading your version of Apigee hybrid requires a supported version of the Kubernetes platform.

The minimum required Kubernetes version is documented in the release notes of the Apigee hybrid release version you are upgrading to.

Confirm that your Kubernetes platform version is supported. If it isn't, follow your platform provider documentation to upgrade it.

Apigee hybrid: supported platforms and versions

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Bootstrapping

- Pull the Apigee hybrid Helm charts from Google Artifact Registry to your local machine.

oci://us-docker.pkg.dev/apigee-release/apigee-hybrid-helm-charts

Use the helm pull command to pull each of the Apigee hybrid component charts:

```
export CHART_REPO=oci://us-docker.pkg.dev/apigee-release/apigee-hybrid-helm-charts
export CHART_VERSION=1.14.1
helm pull $CHART_REPO/apigee-operator --version $CHART_VERSION --untar
helm pull $CHART_REPO/apigee-datastore --version $CHART_VERSION --untar
...
```

- If needed, upgrade cert-manager in your Kubernetes cluster.

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First pull the helm charts for each of the Apigee hybrid components from Google Artifact Registry.

If needed, upgrade the version of cert-manager in your cluster for the version of Apigee hybrid that is being installed.

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Update configuration

- Depending on the version being upgraded to, changes may be necessary in the overrides.yaml configuration file.
- Make a copy of your overrides.yaml file to save the current version.
- Make the corresponding changes as documented in the hybrid release notes documentation for the runtime components.

overrides.yaml

```
...
virtualhosts:
- name: environment-group-name
  selector:
    app: apigee-ingressgateway
    ingress_name: INGRESS_NAME
  sslCertPath: ./certs/CERT_NAME.pem
  sslKeyPath: ./certs/KEY_NAME.key
...
envs:
- name: test
  serviceAccountPaths:
    synchronizer: keypath/sync-sa.json
    udca: keypath/udca-sa.json
    runtime: keypath/runtime-sa.json
...
metrics:
  serviceAccountPath: keypath/metrics-sa.json
udca:
  serviceAccountPath: keypath/udca-sa.json
...
```

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For some newer versions of Apigee hybrid, you may have to update the overrides.yaml file used to configure your runtime plane cluster.

Make a copy of your current overrides file, and save the old file in case you ever need to roll back.

Make the required changes to the overrides file based on the release documentation.

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Apply the upgrade

1. Update the Apigee CRDs in the cluster:

```
$ kubectl apply -k apigee-operator/etc/crds/default/ --server-side --force-conflicts
--validate=false
```

2. Upgrade the Apigee hybrid runtime components, one at a time and in sequence:

```
$ helm upgrade operator apigee-operator/ --install --namespace apigee -f overrides.yaml
$ helm upgrade datastore apigee-datastore/ --install --namespace apigee -f overrides.yaml
$ helm upgrade telemetry apigee-telemetry/ --install --namespace apigee -f overrides.yaml
$ helm upgrade redis apigee-redis/ --install --namespace apigee -f overrides.yaml
$ helm upgrade ingress-manager apigee-ingress-manager/ --install --namespace apigee -f
overrides.yaml
$ helm upgrade {org_name} apigee-org/ --install --namespace apigee -f overrides.yaml
$ helm upgrade {env_release_name} apigee-env/ --install --namespace apigee --set env={env} -f
overrides.yaml
$ helm upgrade {env_group_release_name} apigee-virtualhost/ --install --namespace apigee --set
envgroup={envgroup} -f overrides.yaml
```

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You can now apply the upgrade to your cluster.

First, update the Apigee custom resource definitions in the cluster. Its a best practice to first run the command with the –dry-run flag and validate the command output.

Next, use helm to upgrade the runtime components in the cluster one at a time in sequence. For each upgrade command, you should also use the –atomic flag to rollback the upgrade if there is an error, and the –dry-run=server flag to validate the command first.

After upgrading each helm chart, use kubectl to get the resource and verify it is up and running.

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Delete and update older components

1. Uninstall the old operator release:

   ```
   $ helm uninstall operator -n apigee-system
   ```

2. Delete the apigee-system namespace

   ```
   $ kubectl delete namespace apigee-system
   ```

3. To re-install the deleted cluster-scoped resources, upgrade the operator release in your Apigee namespace:

   ```
   $ helm upgrade operator apigee-operator/ --install
   --namespace apigee --atomic -f overrides.yaml
   ```

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After you have verified all the installations are upgraded successfully, uninstall the older apigee-operator release and delete the apigee-system namespace.

Some cluster-scoped resources are deleted when operator is uninstalled. To re-install them, upgrade operator again in the Apigee namespace.

Make sure to test and validate your API proxies and policies after the upgrade.

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Official documentation

Apigee hybrid official documentation or the release notes may describe additional steps needed for upgrading between specific versions. We strongly recommend that you review this documentation as part of the upgrade planning process.

Upgrading Apigee hybrid:

- Upgrade process
- Apigee hybrid release notes
- Apigee hybrid release process

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In addition to the Apigee hybrid release notes, you should also review the upgrade and release process documentation on the Apigee website.

The documentation may describe additional steps to be performed when upgrading your hybrid release between specific versions.

Upgrade process
Apigee hybrid release notes
Apigee hybrid release process

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Agenda

01	Upgrade Process
02	Rollback Process
03	Quiz

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Let’s discuss the process to roll back Apigee hybrid in the runtime plane.

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Rollback process overview

If an error occurs during or after an upgrade of hybrid, you can roll back the upgrade.

To roll back an upgrade:

- Use the version of the older charts.
- Apply the charts in reverse order starting with apigee-virtualhost through apigee-datastore.
- Re-create the apigee-system namespace.
- Re-apply the apigee-operator chart.
- Revert the custom resource definitions (CRDs).
- Perform cleanup.

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If you observe errors in the Apigee hybrid runtime plane during or after an upgrade, you can roll the upgrade back.

To roll back an upgrade, use the version of the older charts in the reverse order. Start with the apigee-virtualhost chart and work your way back to apigee-datastore.

Then, re-create the apigee-system namespace and update any resource annotations before re-installing the apigee-operator chart into the apigee-system namespace.

Finally re-install the older custom resource definitions, and perform cleanup.

For more details, view the documentation on rolling back to a previous version of Apigee hybrid.

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References

Rollback steps may vary by software version and the target version to roll back to. Before you decide to roll back, it is important to consult the official documentation for details.

The steps to roll back might vary based on the source and target version of the hybrid software distribution.

Consult the rollback process documentation for your version of Apigee hybrid.

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Agenda

01	Upgrade Process
02	Rollback Process
03	Quiz

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Question #1

Question

When upgrading Apigee hybrid:

A. You may need to upgrade the underlying Kubernetes platform.

B. You do not have to back up the runtime datastore used in your current Apigee hybrid installation.

C. You can upgrade Apigee hybrid from any previous version to the latest released version.

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Question #1

Answer

When upgrading Apigee hybrid:

A. You may need to upgrade the underlying Kubernetes platform.

B. You do not have to back up the runtime datastore used in your current Apigee hybrid installation.

C. You can upgrade Apigee hybrid from any previous version to the latest released version.

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Explanation:

A. Correct! The underlying Kubernetes platform may need to be upgraded to meet the version requirement of the Apigee hybrid software that is being upgraded.

B. Incorrect. It is a recommended best practice to always back up the runtime datastore used in your current Apigee hybrid installation.

C. Incorrect. Depending on the version of your current hybrid installation, you may have to first upgrade to an intermediate version before you upgrade to the latest released version of Apigee hybrid.

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Question #2

Question

When upgrading Apigee hybrid in a production environment, what sequence should you follow to upgrade the runtime plane component charts in the cluster?

A. Upgrade redis, ingress-manager, virtualhosts, telemetry, organization, environments, operator, and datastore, in that order.

B. Upgrade operator, datastore, telemetry, redis, ingress-manager, organization, environments, and virtualhosts, in that order.

C. Upgrade organization, datastore, telemetry, redis, ingress-manager, virtualhosts, environments, and operator, in that order.

D. Upgrade operator, environments, organization, datastore, redis, ingress-manager, virtualhosts, and telemetry, in that order.

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Question #2

Answer

When upgrading Apigee hybrid in a production environment, what sequence should you follow to upgrade the runtime plane component charts in the cluster?

A. Upgrade redis, ingress-manager, virtualhosts, telemetry, organization, environments, operator, and datastore, in that order.

B. Upgrade operator, datastore, telemetry, redis, ingress-manager, organization, environments, and virtualhosts, in that order.

C. Upgrade organization, datastore, telemetry, redis, ingress-manager, virtualhosts, environments, and operator, in that order.

D. Upgrade operator, environments, organization, datastore, redis, ingress-manager, virtualhosts, and telemetry, in that order.

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Explanation:

A. Incorrect. When upgrading the hybrid runtime component charts, you should upgrade the operator chart first, then the datastore, telemetry, redis, and ingress-manager charts, followed by the organization, environments, and virtualhosts charts in that order.

B. Correct! This is the correct sequence to upgrade the hybrid runtime components in the cluster. Upgrade the operator chart first, then the datastore, telemetry, redis, and ingress-manager charts, followed by the organization, environments, and virtualhosts charts in that order.

C. Incorrect. When upgrading the hybrid runtime component charts, you should upgrade the operator chart first, then the datastore, telemetry, redis, and ingress-manager charts, followed by the organization, environments, and virtualhosts charts in that order.

D. Incorrect. When upgrading the hybrid runtime component charts, you should upgrade the operator chart first, then the datastore, telemetry, redis, and ingress-manager charts, followed by the organization, environments, and virtualhosts charts in that order.

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Question #3

Question

What are some steps to roll back an upgrade of Apigee hybrid?

A. Re-install the previous custom resource definitions.

B. Apply the previous version of the hybrid helm charts in the same order as they were applied during the upgrade.

C. Re-create the apigee-system namespace.

D. Re-apply the apigee-operator helm chart.

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Question #3

Answer

What are some steps to roll back an upgrade of Apigee hybrid?

A. Re-install the previous custom resource definitions.

B. Apply the previous version of the hybrid helm charts in the same order as they were applied during the upgrade.

C. Re-create the apigee-system namespace.

D. Re-apply the apigee-operator helm chart.

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Explanation:

A. Correct! The CRDs installed in the cluster must be reverted to their previous versions.

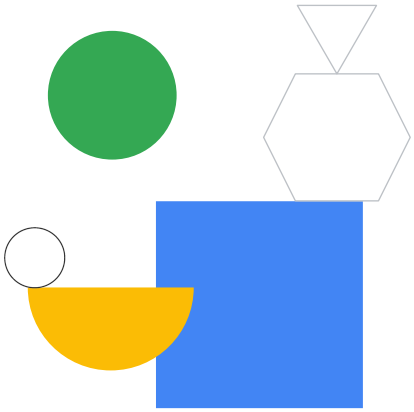
B. Incorrect. The previous version of the hybrid helm charts must be applied in reverse order starting with the apigee-virtualhost through apigee-operator.

C. Correct! The apigee-system namespace must be re-created.

D. Correct! The apigee-operator chart must be applied to the cluster to re-install the operator resource in the apigee-system namespace.

Google Cloud

Review: Upgrade



In this module, you learned how to upgrade the Apigee hybrid runtime plane to a newer version of the software distribution.

You also learned about the rollback process in case you need to roll back Apigee hybrid due to errors in the newer version.

In the next module, you will learn how Apigee hybrid logs system informational and error messages and how you can use analytics and metrics data to monitor and troubleshoot Apigee hybrid.